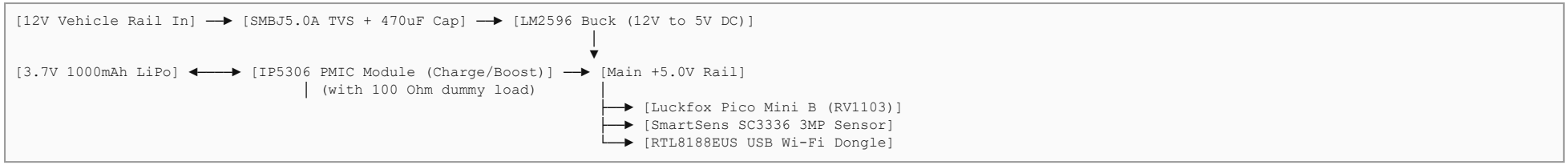
ELECTRICAL POWER BUDGET & THERMAL CALCULATION (SHEET 1)

1. ELECTRICAL POWER DISTRIBUTION ARCHITECTURE



2. SUBSYSTEM CURRENT SCHEDULE (MEASURED AT 5.0V DC SYSTEM RAIL)

LOAD NODE / SUBSYSTEM	STANDBY CURRENT (MA)	ACTIVE STREAMING (MA)	PEAK TRANSIENT (MA)	OPERATING VOLTAGE	ACTIVE POWER (W)
RV1103 ARM Cortex-A7 Core	50.0	120.0	180.0	5.0V DC	0.60 W
Hardware ISP 3.2 Engine	0.0	70.0	90.0	5.0V DC	0.35 W
Hardware H.264 VPU Encoder	0.0	50.0	70.0	5.0V DC	0.25 W
64MB DDR2 Memory Array	15.0	30.0	45.0	5.0V DC	0.15 W
SmartSens SC3336 3MP Sensor	0.0	75.0	110.0	5.0V DC	0.38 W
RTL8188EUS USB Wi-Fi Adapter	30.0	130.0	250.0	5.0V DC	0.65 W
IP5306 Keep-Alive Load (R1: 100R)	50.0	50.0	50.0	5.0V DC	0.25 W
Status Optical Indicator (R4: 470R)	0.0	7.0	10.0	5.0V DC	0.035 W
TOTAL 5.0V RAIL LOAD:	**145.0 mA**	**532.0 mA**	**805.0 mA**	**5.0V DC**	**2.66 Watts**

	ORGANIZATION / PROJECT: DASHCAM PROTOTYPE PROJECT	SIZE: A4	DOCUMENT NUMBER: ENG-DASH-PWR-001	DATE: 2026-10-08	SHEET: 1 OF 2
			DESIGN ENGINEER: Souvik H	ENGINEER REGISTRATION NO.: IN-ENG-2022-22824	
	DOCUMENT TITLE: ELECTRICAL POWER & THERMAL SPEC	REV: 1.4	ENGINEERING DISCIPLINE: POWER ELECTRONICS & THERMAL ENGR	STATUS: RELEASED	

ELECTRICAL POWER BUDGET & THERMAL CALCULATION (SHEET 2)

3. BATTERY CAPACITY & OPERATIONAL AUTONOMY CALCULATION

CALCULATION STEP	FORMULA / TECHNICAL PARAMETER	RESULTING VALUE
Total Stored Energy	Battery Capacity (1.0 Ah) x Nominal Voltage (3.7V DC)	3.70 Watt-hours (Wh)
Effective Usable Energy	3.70 Wh x IP5306 Boost Conversion Efficiency (eta = 90.0%)	3.33 Watt-hours (Wh)
Active Streaming Autonomy	Usable Energy (3.33 Wh) / Continuous System Power (2.66 W)	**1.25 Hours (75 Minutes continuous streaming)**
Standby State Autonomy	Usable Energy (3.33 Wh) / Quiescent Standby Power (0.725 W)	**4.59 Hours (~275 Minutes)**

4. VEHICLE IN-LINE CHARGING & THERMAL DISSIPATION ANALYSIS

- **Charging Dynamics:** LM2596 delivers up to 2.0A at 5.0V. System draws 532mA; remaining ~1.0A charges LiPo to 100% in ~65 min.
- **Thermal Dissipation:** RV1103 SoC dissipates ~0.8W. Requires a 9x9x5mm extruded aluminum heatsink.
- **Automotive Environment:** Windshield temperatures can exceed 65 Deg C. LiPo charging must be inhibited above 45 Deg C ambient.

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-PWR-001	DATE 2026-10-08	SHEET 2 OF 2	
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